

Project Initialization and Planning Phase

Team ID: SWTID-2026-7446

Project Title: ComicCraft - AI Comic Story Creator using Gemini Models

Project Proposal (Proposed Solution) Report

Project Overview

Objective: To create a web-based application that uses AI to generate personalized comic book stories and illustrations from user-provided prompts.

Scope: The project covers prompt-based comic outline generation, narration and dialogue creation, comic-style image generation, panel organization, web preview, and PDF export.

Problem Statement

Description: Comic creation can require substantial creative and technical effort across story planning, dialogue, illustration, layout, and export. ComicCraft addresses this by automating these stages through an integrated AI workflow.

Impact: The application makes comic creation accessible to non-artists, storytellers, and hobbyists while providing a seamless flow from a simple story idea to a complete downloadable comic.

Proposed Solution

Approach: ComicCraft combines FastAPI, Google Gemini AI models, and Stable Diffusion to generate structured comic storylines, detailed narration and dialogues, and comic-style illustrations.

Key Features:

- Gemini Flash generates a structured 5-panel comic outline.
- Gemini Pro expands the outline into detailed narration and character dialogues.
- Stable Diffusion generates comic-style illustrations from image prompts.
- The backend organizes panels into a structured layout and exports the comic as a multi-page PDF.
- A web interface collects story prompt, character name, setting, tone, and art style.
- Users can preview the generated comic and download it as a PDF.

Project Workflow

Milestone 1: Model selection and Architecture

Activity 1.1: Research and select the appropriate generative AI model.

Activity 1.2: Define the application architecture.

Activity 1.3: Set up the development environment.

Milestone 2: Core functionalities Development

Activity 2.1: Develop core functionalities for outline, story, image, layout, and PDF generation.

Activity 2.2: Implement the FastAPI backend to manage routing and user input processing.

Milestone 3: routes.py Development

Activity 3.1: Write the main application logic and core routes.

Milestone 4: Frontend Development

Activity 4.1: Design and develop the user interface.

Activity 4.2: Create dynamic templates with FastAPI's Jinja2.

Milestone 5: Deployment

Activity 5.1: Prepare the application for local deployment.

Activity 5.2: Test and verify local deployment.

Technical Architecture

Frontend: HTML, CSS, and Jinja2 templates for input collection and comic preview.

Backend: FastAPI for server-side operations, routing, AI workflow coordination, layout building, and PDF export.

AI Integration: Google Gemini APIs and Hugging Face Diffusers / Stable Diffusion.

AI flow: Gemini Flash → structured 5-panel outline; Gemini Pro → narration and dialogue; Stable Diffusion → panel illustrations.

Resource Requirements

Software: Python, FastAPI, Uvicorn, Jinja2, python-multipart, google-generativeai, diffusers, transformers, fpdf, Pillow, accelerate, PyTorch.

Development Environment: Jupyter Notebook / PyCharm as stated in the project documentation.

APIs/Services: Google Gemini API and Hugging Face Diffusers / Stable Diffusion.

Implementation Details

Core functions: generate_outline(), generate_story(), generate_image(), build_comic_layout(), and save_pdf().

Routes: /, /generate, /generate-comic/json, /export-success, and /test-image.

PDF Export: exporters.py compiles panel images and narration into a multi-page PDF using FPDF.

Deployment: The documented local server command is ``uvicorn app.main:app --reload``, with the application accessed at the local FastAPI address.

Conclusion

ComicCraft demonstrates an end-to-end AI-assisted comic generation workflow, combining Gemini Flash and Gemini Pro for storytelling with Stable Diffusion for illustrations. It supports personalized story generation, visual comic creation, preview, and downloadable PDF export.